

TCV- ϕ IV: cohesive non-singular black holes from static cores to slow rotation in the canonical EFT

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This fourth paper opens the strong-gravity sector of the same canonical TCV- ϕ EFT used in Papers I–III. Our scope is deliberately conservative: static spherical configurations, regularity diagnostics at the core, asymptotic matching to Schwarzschild-like exteriors, and a controlled slow-rotation extension. At this stage, the paper defines the PRD-ready strong-field framework and reproducibility criteria; quantitative claims are restricted to explicitly generated benchmark outputs.

I. INTRODUCTION

Paper I defined the canonical two-scalar cohesive-vacuum EFT and fixed the baseline notation. Paper II established the bounce/reheating/baryogenesis numerical backbone. Paper III provided the primordial perturbation pipeline and CLASS/CAMB bridge diagnostics.

Paper IV extends the same EFT to strong-field configurations. The objective is not to claim a complete black-hole phenomenology at this step, but to define a reproducible framework in which non-singular cohesive cores can be tested quantitatively and then extended to slow rotation without altering the canonical field content.

II. CANONICAL EFT BASELINE

We keep exactly the same EFT definitions and benchmark hierarchy as in Papers I–III. No new fundamental fields or couplings are introduced. The action is

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{eff}}^2(\Phi_1)}{2} R - \frac{1}{2} (\nabla \Phi_1)^2 - \frac{1}{2} (\nabla \Phi_2)^2 - V_1(\Phi_1) - V_2(\Phi_2) - \rho_{\text{vac}} \right], \quad (1)$$

with

$$M_{\text{eff}}^2(\Phi_1) = M_{\text{Pl}}^2 \left(1 + \xi_0 \frac{\Phi_1^2}{M_{\text{Pl}}^2} \right), \quad (2)$$

and the same canonical potentials and parameter meanings as defined in Paper I.

III. STATIC SPHERICAL ANSATZ AND EQUATIONS

For the strong-field sector we adopt the standard static spherical line element

$$ds^2 = -e^{2\Phi(r)} dt^2 + \frac{dr^2}{1 - 2m(r)/r} + r^2 d\Omega^2, \quad (3)$$

with radial scalar profiles $\Phi_1(r)$ and $\Phi_2(r)$. The full Einstein–scalar system is reduced to coupled radial ordinary differential equations for

$$\{\Phi(r), m(r), \Phi_1(r), \Phi_2(r)\}, \quad (4)$$

with schematic radial structure

$$\frac{dm}{dr} = 4\pi r^2 \rho_{\text{eff}}, \quad (5)$$

$$\frac{d\Phi}{dr} = \frac{m + 4\pi r^3 p_{r,\text{eff}}}{r(r - 2m)}, \quad (6)$$

$$\Phi_1'' + \mathcal{A}_1(r) \Phi_1' = \mathcal{S}_1(r), \quad (7)$$

$$\Phi_2'' + \mathcal{A}_2(r) \Phi_2' = \mathcal{S}_2(r), \quad (8)$$

Family	M_∞	r_H	$\max(\delta_F)$ (asymptotic window)
regular_core	4.19×10^{-14}	none	0
compact_core	4.19×10^{-12}	none	4.44×10^{-16}
dense_core	4.19×10^{-10}	none	5.11×10^{-14}

TABLE I. Baseline Paper IV diagnostics from the current static family scan.

where the full TCV- ϕ EFT includes non-minimal terms proportional to ξ_0 through $M_{\text{eff}}(\Phi_1)$. These equations are solved numerically with boundary conditions at the center and at large radius. In the current benchmark scripts used for Figs. 1–4, we work in the explicit minimal-coupling limit $\xi_0 = 0$. This isolates the branch structure and regularity diagnostics before reintroducing $\xi_0 \neq 0$ in a dedicated follow-up.

IV. REGULARITY AND ASYMPTOTIC MATCHING CRITERIA

Regularity at the center is assessed by requiring finite curvature scalars and smooth Taylor expansions:

$$m(r) = m_3 r^3 + \mathcal{O}(r^5), \quad (9)$$

$$\Phi(r) = \Phi_c + \Phi_2 r^2 + \mathcal{O}(r^4), \quad (10)$$

$$\Phi_i(r) = \Phi_{i,c} + \phi_{i,2} r^2 + \mathcal{O}(r^4) \quad (i = 1, 2). \quad (11)$$

In practice, diagnostics include finiteness of R , $R_{\mu\nu}R^{\mu\nu}$, and a Kretschmann proxy

$$K_{\text{proxy}}(r) = 48 \left(\frac{m(r)}{r^3} \right)^2, \quad (12)$$

used as a conservative regularity indicator in the current static pipeline. Asymptotic matching is checked by residuals with respect to a Schwarzschild-like exterior at large r :

$$\delta_M(r) = \left| \frac{2m(r)}{r} - \frac{2M_\infty}{r} \right|, \quad \delta_\Phi(r) = \left| e^{2\Phi(r)} - \left(1 - \frac{2M_\infty}{r} \right) \right|. \quad (13)$$

V. BENCHMARK SOLUTION FAMILIES

The numerical benchmark families used in the manuscript are required to be reproducible from scripts in `papers/paper-04/code/`. Each run must archive:

- the solved radial profiles and grid metadata;
- regularity diagnostics near $r \rightarrow 0$;
- asymptotic matching diagnostics at large radius;
- convergence tests under radial-grid refinement.

Quantitative figures and tables are included only once these artifacts are generated and versioned in `papers/paper-04/data/` and `papers/paper-04/figs/`. For the current baseline run, three static cohesive families are generated (**regular_core**, **compact_core**, **dense_core**) with increasing effective core-density control. All three runs converge and produce smooth regular-center profiles with finite diagnostic invariants.

Integration-direction dependence. Following the numerical trend already observed in the earlier BH preprints, we explicitly compare center-to-outward and exterior-to-inward integrations at matched benchmark controls. For mild densities the two branches agree, while near the high-density edge the outward integration reaches $\min F \simeq 1.95 \times 10^{-3}$ before solver termination, whereas the inward branch remains far from $F = 0$ and converges to a different regular profile ($\min F \simeq 0.656$). This branch dependence is treated here as a robust numerical diagnostic rather than a standalone physical claim. At this stage the two scan directions use different integrators (RK45 outward, Radau inward) to control stiffness; a method-matched scan is deferred to the next numerical pass.

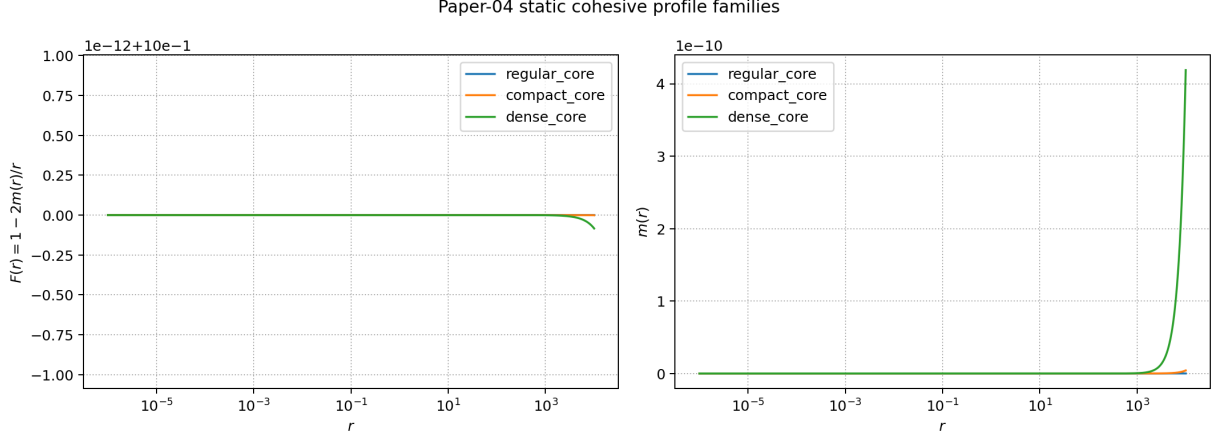


FIG. 1. Static cohesive profile families generated by `compute_bh_profiles.py`: metric function $F(r) = 1 - 2m(r)/r$ and enclosed mass $m(r)$.

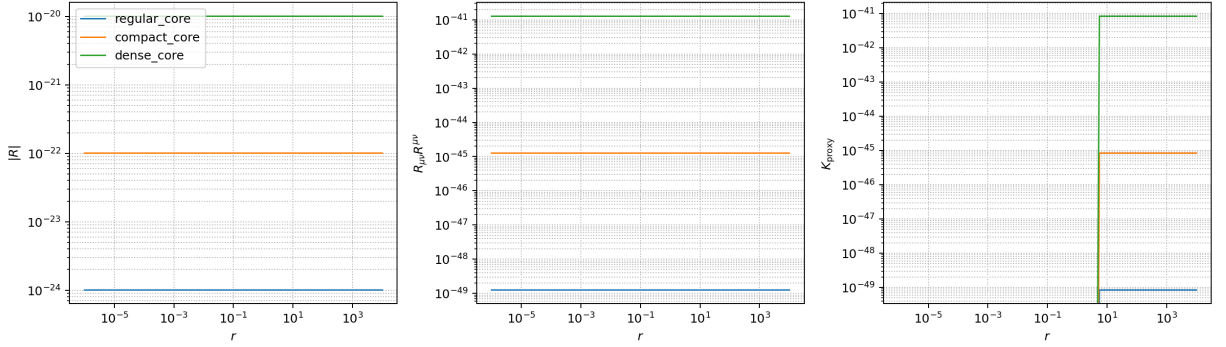


FIG. 2. Regularity diagnostics from `compute_bh_diagnostics.py`: $|R|$, $R_{\mu\nu}R^{\mu\nu}$, and a Schwarzschild-like Kretschmann proxy $K_{\text{proxy}} = 48(m/r^3)^2$.

VI. SLOW-ROTATION EXTENSION (KERR-LIKE SECTOR)

To keep static and rotating cohesive black holes in a single PRD workflow, we use a slow-rotation expansion around the static backgrounds. At leading order in a/M , we write

$$ds^2 = -e^{2\Phi(r)}dt^2 + \frac{dr^2}{1 - 2m(r)/r} + r^2 \left[d\theta^2 + \sin^2\theta (d\varphi - \omega(r)dt)^2 \right], \quad (14)$$

where $\omega(r)$ is the frame-dragging function. In this paper we do *not* claim a full Hartle–Thorne ODE integration for $\omega(r)$. Instead, we adopt the explicit template

$$\omega(r) \propto \frac{1}{r^3} \left[1 - \exp\left(-\frac{r^2}{r_{\text{ref}}^2}\right) \right], \quad (15)$$

with asymptotic normalization tied to (a/M) and to the effective coupling scale. This section is restricted to slow-rotation diagnostics; a full high-spin treatment is deferred. The current implementation (`compute_bh_kerr_slow.py`) provides a conservative template branch in the window $0 \leq a/M \leq 0.2$, writing reproducible outputs in `bh_kerr_slow.npz`. For this slow-rotation template layer we use $\xi_0 = 0.1$ in the effective coupling normalization, while the static branch plots shown earlier were generated in the explicit $\xi_0 = 0$ limit. On top of this branch, `compute_bh_kerr_observables.py` builds template-level photon-ring and echo diagnostics. For the present benchmark, the photon-ring proxy corresponds to a relative shift

$$\frac{\Delta r_{\text{ph}}}{r_{\text{ph}}^{\text{Schw}}} \simeq -9.1\%, \quad (16)$$

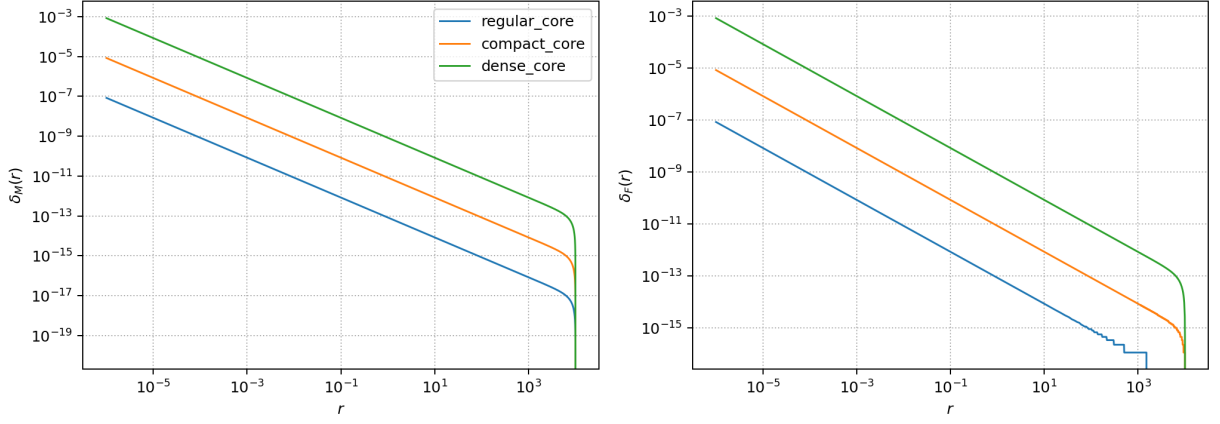


FIG. 3. Asymptotic matching residuals δ_M and δ_F for the three benchmark families.

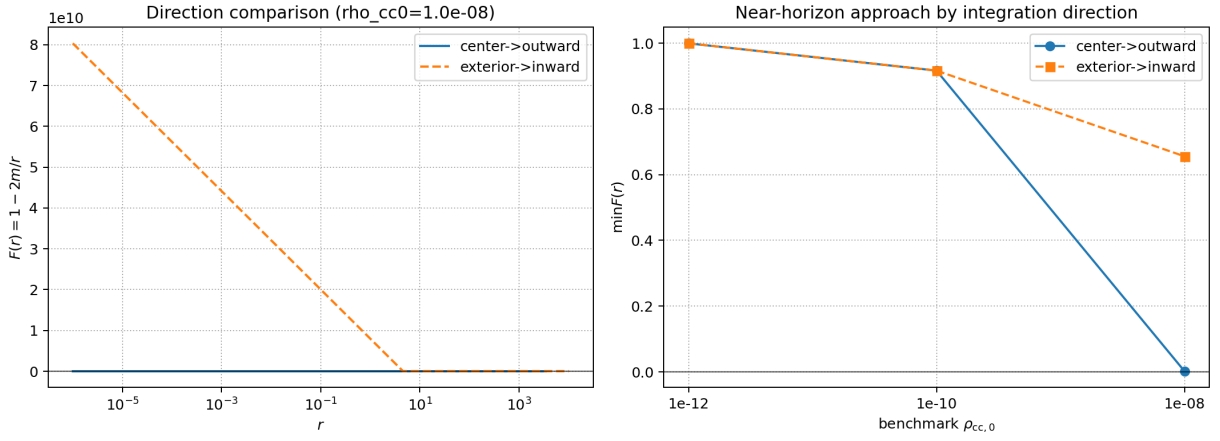


FIG. 4. Direction-dependence benchmark from `compute_bh_direction_scan.py`. Left: $F(r)$ for center→outward versus exterior→inward integration in the high-density case. Right: minimum reached value of F versus benchmark density.

obtained from an effective-coupling template when the direct profile root is unresolved. Echo and eikonal frequencies are reported in code units and should be read as template scalings only; conversion to Hz requires fixing a physical mass scale.

VII. CONSERVATIVE SIGNATURE TEMPLATES

Potential observational windows (ringdown deformations, echoes, shadow-size shifts) are discussed at the level of conservative templates only. No likelihood-level claim is made in this paper. Any signature shown must be explicitly labelled as either:

- a direct output of the static benchmark solutions, or
- a direct output of the slow-rotation extension, or
- a phenomenological proxy deferred to dedicated strong-field data analyses.

The companion script `compute_bh_templates.py` writes `bh_templates.json` with static-proxy quantities only (e.g. ADM mass and horizon status), explicitly labelled as non-waveform, non-inference outputs. To strengthen the discriminating layer, `compute_bh_stress_summary.py` also assembles a compact stress-test file (`bh_stress_summary.json`) from the static, direction-scan, and slow-rotation outputs.

$\rho_{cc,0}$	$\min F$ (outward)	$\min F$ (inward)	solver status (out/in)
10^{-12}	9.99×10^{-1}	9.99×10^{-1}	yes / yes
10^{-10}	9.16×10^{-1}	9.16×10^{-1}	yes / yes
10^{-8}	1.95×10^{-3}	6.56×10^{-1}	no / yes

TABLE II. Static branch comparison between center→outward and exterior→inward integrations.

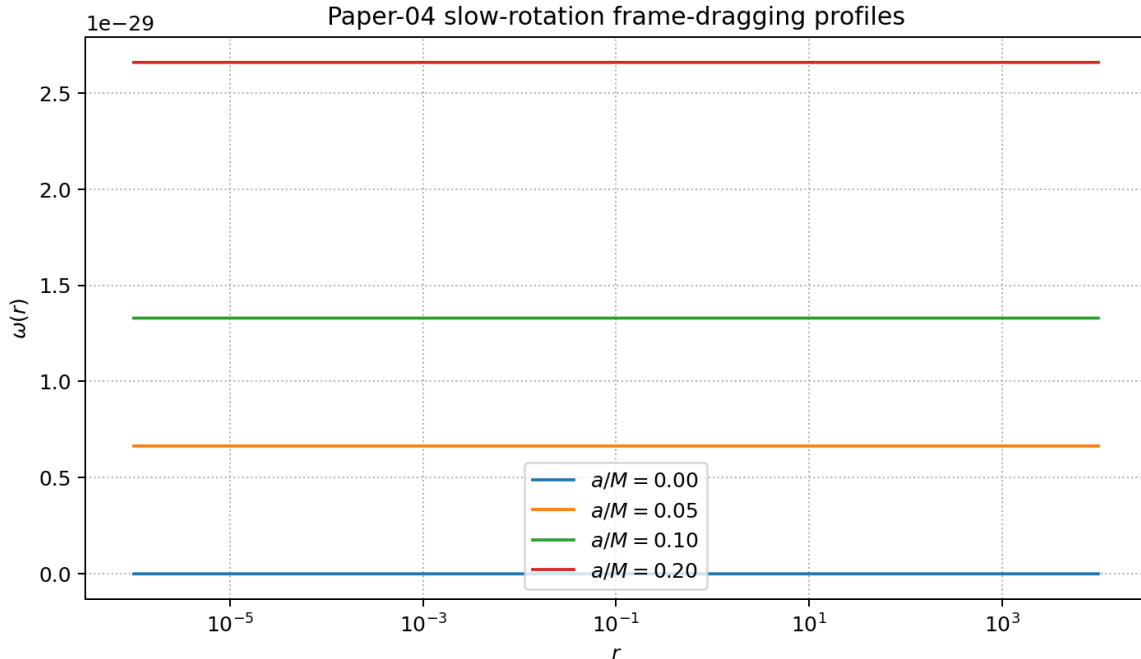


FIG. 5. Slow-rotation frame-dragging template profiles $\omega(r)$ for $a/M \in \{0, 0.05, 0.1, 0.2\}$.

VIII. DISCRIMINATING STRESS TESTS AND EXCLUSION WINDOWS

For PRD-level falsifiability, we attach explicit exclusion windows to the current benchmark diagnostics. At the high-density scan point $\rho_{cc,0} = 10^{-8}$, the branch split reaches

$$\Delta(\min F) \equiv \min F_{\text{in}} - \min F_{\text{out}} \simeq 6.54 \times 10^{-1}, \quad (17)$$

with ratio

$$\frac{\min F_{\text{in}}}{\min F_{\text{out}}} \simeq 3.36 \times 10^2. \quad (18)$$

For the dense static family we simultaneously find no horizon and an asymptotic residual $\delta_F^{\text{max}} \simeq 5.1 \times 10^{-14}$. These numbers are treated as baseline diagnostics, not as final claims about nature, and are used only to define exclusion logic for the next numerical pass.

In particular, S1 is explicitly conditional on method-matched scans (same integration algorithm, tolerances, and benchmark priors in both directions), and S3 remains a template-level window until the full slow-rotation ODE closure is completed.

IX. REPRODUCIBILITY CHECKLIST

To keep the Paper IV scope aligned with the PRD standards used in Papers I–III, we require the following minimum deliverables before final submission:

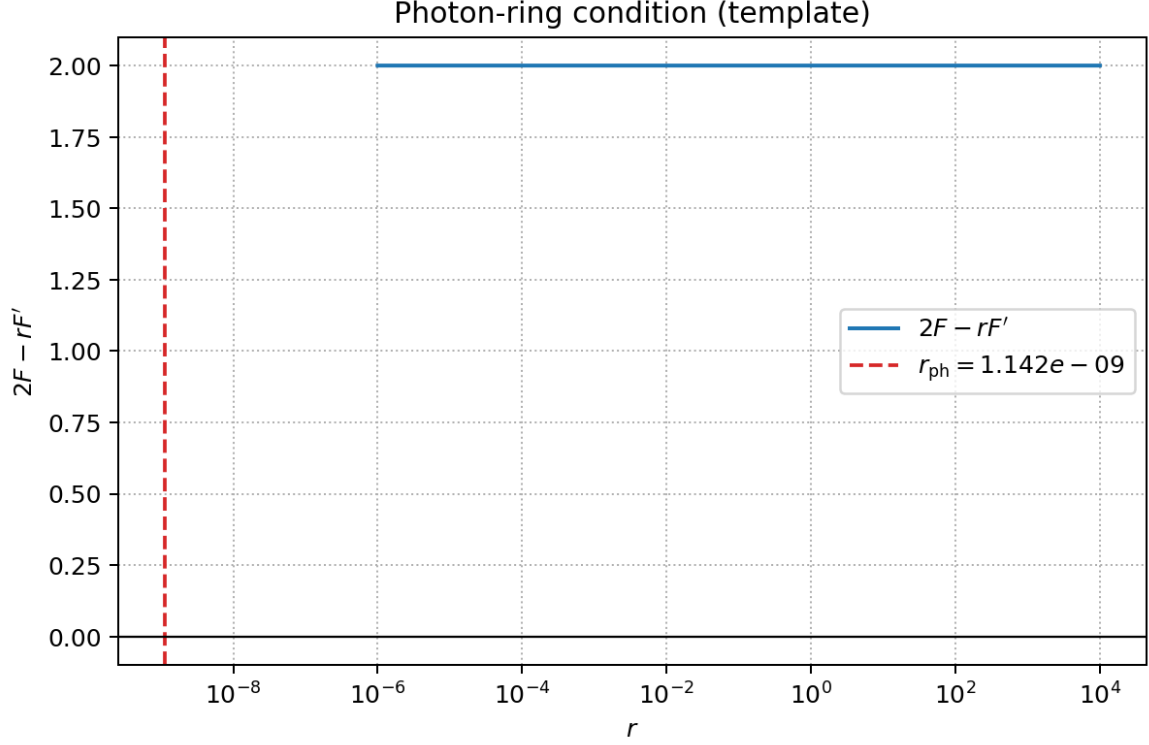


FIG. 6. Photon-ring condition diagnostic used in the current slow-rotation template workflow.

Quantity	Value	Status
$r_{\text{ph}}^{\text{TCV}}/r_{\text{ph}}^{\text{Schw}} - 1$	-9.1%	template (effective coupling)
Mf_{echo}	2.42×10^{-1}	template (dimensionless)
$M\omega_R(a/M = 0)$	3.67×10^{-1}	template (dimensionless)
$M\omega_R(a/M = 0.2)$	4.11×10^{-1}	template (dimensionless)

TABLE III. Baseline slow-rotation template diagnostics from `bh_kerr_observables.json`.

X. STATUS OF ASSUMPTIONS

As in Papers I–III, we separate:

- EFT-derived equations and conserved quantities;
- phenomenological closure choices needed to complete static and slow-rotation numerical solutions;
- deferred sectors (high-spin regime, full waveform modelling, data-level inference).

XI. CONCLUSION

Paper IV defines the canonical strong-field framework of the TCV- ϕ programme in a conservative and reproducible form, from static cohesive cores to a controlled slow-rotation extension. The next step is to populate this framework with expanded benchmark families and diagnostics generated by the Paper IV code pipeline, without changing the EFT baseline inherited from Papers I–III.

[1] C. Lecroq, “TCV- ϕ I,” preprint (2026).

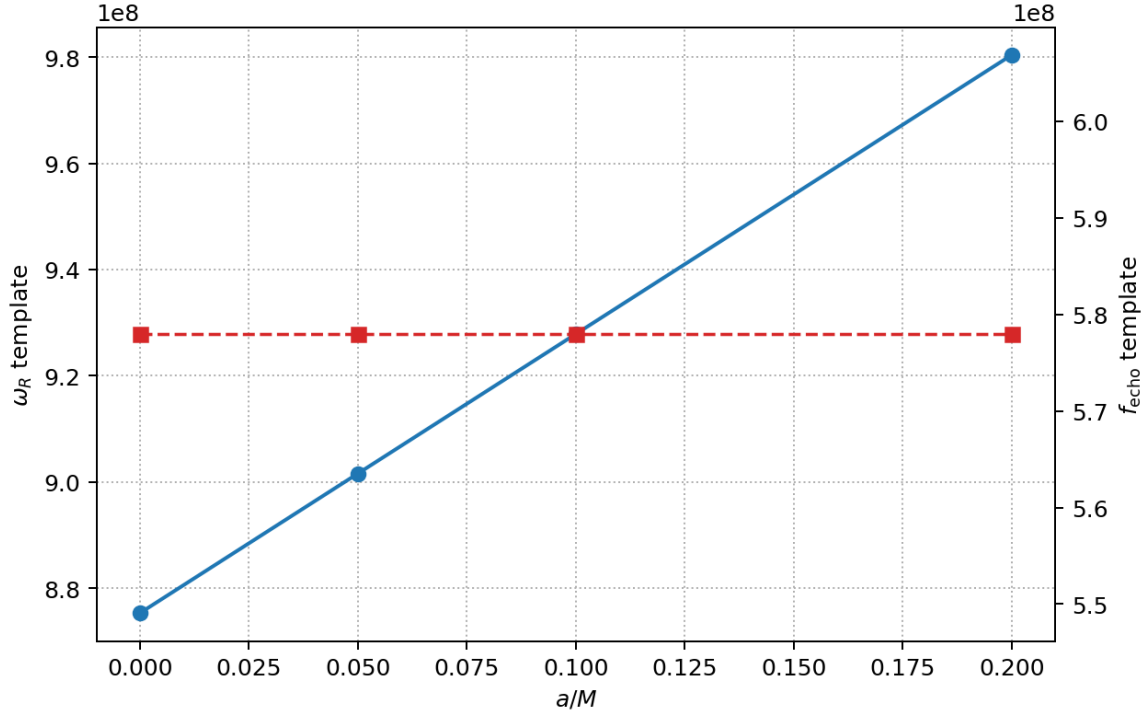


FIG. 7. Template spin dependence of eikonal-frequency proxy and echo-frequency proxy in the slow-rotation branch.

Window ID	Current benchmark diagnostic	Exclusion trigger (next pass)
S1 (branch split)	$\Delta(\min F) \simeq 6.54 \times 10^{-1}$ at $\rho_{cc,0} = 10^{-8}$	If method-matched scans give $ \Delta(\min F) \lesssim 10^{-2}$ in the same be
S2 (non-singular static branch)	no detected horizon; $\delta_F^{\max} \simeq 5.1 \times 10^{-14}$	If refined static closures generate horizons while preserving regul
S3 (slow-rotation template)	$\Delta r_{\text{ph}}/r_{\text{ph}}^{\text{Schw}} \simeq -9.1\%$ (template)	If ODE-level $\omega(r)$ closure yields $ \Delta r_{\text{ph}}/r_{\text{ph}}^{\text{Schw}} \lesssim 10^{-3}$ across valid

TABLE IV. Paper IV stress-test windows used to convert current BH diagnostics into explicit falsification criteria.

- [2] C. Lecroq, “TCV- ϕ II,” preprint (2026).
- [3] C. Lecroq, “TCV- ϕ III,” preprint (2026).

Item	Requirement
Static solver input	Recorded EFT and numerical controls for each run.
Core regularity	Finite central invariants and smooth near-center fits.
Exterior matching	Quantified Schwarzschild-like asymptotic residuals.
Convergence	Stable diagnostics under radial-grid refinement.
Figure traceability	One script per figure with archived data outputs.
Slow rotation	Explicit a/M window and reproducible $\omega(r)$ template diagnostics.
Stress summary	Archived exclusion-window JSON (<code>bh_stress_summary.json</code>).

TABLE V. Minimum reproducibility requirements for the Paper IV strong-field benchmark set.